

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 1

Year: 2019

Subject: 392133 Physics III

Section: 15-18

Date: 10 October 2019

Time: 08.00-10.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is composed into 1 section. There will be 7 pages (including this page) and they are worth 60 points (10 problems).

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators are NOT allowed in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

1. [5 points] An object is placed 15 cm in front of a concave mirror with a focal length of 30 cm. What is the magnification?

2. [5 points] A plane convex lens is made of glass ($n = 1.5$) with one flat surface and the other having a radius of 20 cm. What is the focal length (in cm) of the lens?

3. [5 points] An object 50 cm high is placed 1.0 m in front of a converging lens whose focal length is 1.5 m. Determine the image height (in cm).

4. [5 points] A far-sighted student has a near point of 50 cm. Calculate the focal length (in cm) of the glasses needed so the near point will be normal (25 cm).

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5. [10 points] An object is placed 25 cm in front of a lens of focal length 20 cm. 150 cm past the first lens is a second lens of focal length 25 cm. What is the resulting magnification of the object in this setup?

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6. [5 points] Estimate the distance (in cm) between the central bright region and the third dark fringe on a screen 5.00 m from two double slits 0.500 mm apart illuminated by 500 nm light.

7. [5 points] A soap bubble ($n = 1.35$) is floating in air. If the thickness of the bubble wall is 100 nm, what visible light wavelength (400-700 nm) is most strongly reflected?

8. [5 points] How wide must a narrow slit be if the first diffraction minimum occurs at $\pm 30^\circ$ with laser light of 540 nm?

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9. [5 points] A laser ($\lambda = 540$ nm) is used to calibrate a diffraction grating. If the first-order maximum occurs at 30° , what is the line spacing, d ?

10. [10 points] Unpolarized light is passed through two successive Polaroid filters. The second filter has the transmission axis at 45.0° to the first filter. What percentage of light gets through?

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Asst.Prof.Dr.Amarin Ratanavis

Dr. Thanakorn Iamsasri